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Regional cerebral glucose metabolism in man during wakefulness, sleep, and dreaming

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Regional cerebral glucose metabolism was measured by positron emission tomography in 4 healthy male volunteers, both during wakefulness and sleep, using the 2-deoxy-D-[2-^{18}F]glucose method. While 3 of the subjects did not report dreaming and, during sleep stages I–IV of variable duration, exhibited a non-selective decrease in metabolic rates averaging 12.6 ± 4.73% (mean ± S.D.) for the entire brain, the fourth volunteer who experienced an extended nightmare during his sleep examination showed a generalized increase in cerebral glucose utilization ranging from 2.1% in lentiform nucleus to 30.0% in the superior frontal cortex, with a weighted whole brain average of 16.4%. These findings suggest that energy metabolism in the human brain is generally depressed during slow-wave sleep as opposed to a — possibly differential — activation during dreaming.

A fairly consistent pattern of sleep-associated changes in spontaneous or evoked neurophysiological activity of various brain regions has been observed in a number of human and animal studies. Despite the close coupling between neuronal function and cerebral energy metabolism[12]; however, reports on regional metabolic changes in the brains of sleeping animals are contradictory[5,7,9] and, applying the Kety-Schmidt method, no significant difference in global cerebral oxidative metabolism was found between sleep and wakefulness in human subjects[6]. Therefore, in the present study positron emission tomography (PET) of 2-deoxy-D-[2-^{18}F]glucose (FDG) affording non-invasive repeatable quantification of local cerebral glucose utilization[10] was employed to determine possible differential effects of sleep, with and without dreaming, on regional brain metabolism of normal volunteers also measured during wakefulness. In view of the exploratory nature of this investigation, the apparent mismatch between comparatively frequent changes in sleep state and lengthy PET recordings basically requiring steady state conditions, as well as for methodological reasons, precise definition of the sleep stages reflected by the metabolic images was not attempted and neurophysiological sleep monitoring was restricted to an absolute minimum.

The study was carried out in 6 healthy male medical doctors aged 28–38 years, all neurological residents familiar with the technical procedure. Because of the logistics involved in providing FDG examinations were performed around noon. Each volunteer was studied twice on different days, following approx. 28 h on duty with little or no sleep at all. Two of them were excluded from further analysis though, as they did not remain fully asleep during the first 30 min of their sleep examination. Sleep and waking studies were performed in random order. One to two hours before FDG injection only 3 active EEG electrodes (frontal, centrotemporal, parieto-occipital) were attached on each side, and a grounding electrode on the forehead, for sleep monitoring. (No electro-oculogram was written in order to avoid artifactual attenuation of the emitted radiation by metallic electrodes and leads in the field of view of the PET detectors.) Subjects were then placed on a reclining chair, and short teflon catheters were inserted into one antecubital vein for injection, as well as into a

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contralateral dorsal hand vein for blood sampling. That hand was subsequently held in a temperature-controlled water bath at 44 °C[8], while the volunteer was resting in a comfortable supine position with his eyes closed and his head inside the scanner gantry. During waking studies, wakefulness was ascertained by continuous direct observation and occasional prompting of faint finger movements. Following uninterrupted sleep examination, subjects were interviewed as to their dreaming experience during the study: 3 volunteers did not recall dreaming, while the fourth one reported an extended frightening dream with intimidating human figures moving around and towards him.

Well before the beginning of each examination lights in the PET room were dimmed and ambient noise was minimal all the time. Sleep studies were not initiated unless at least sleep stage II was documented by EEG. The PET procedure then followed as described elsewhere[3]. In brief, approx. 5 mCi FDG (specific activity = 10–20 mCi/mg) in 3–5 ml normal saline solution were injected as a rapid intravenous bolus, concurrently the four ring/seven slice positron camera (Scanditronix PC 384, in-plane full width at half-maximum = 8 mm)[1] was started, and dynamic recordings were obtained for 30 min.

During two subsequent periods of 10 min duration each, the entire brain was scanned from base to vertex in transaxial planes (average thickness = 11 mm) yielding a minimum of 9 equally spaced activity images, which were then transformed into metabolic maps using the plasma activity–time curve estimated from individual blood samples, and a modified operational equation[8] of the Sokoloff model. On these images an average of 210 standardized regions of interest per case were outlined employing a user-writ-

TABLE I

Regional cerebral metabolic rate for glucose in normal human subjects during sleep and wakefulness (μmol/100 g/min)

Region	*Non-dreamers (n = 3)*		*Dreamer*	
	Awake (mean ± S.D.)	*% Change asleep (mean ± S.D.)*	*Awake*	*% Change asleep*
Orbital cortex	42.1 ± 1.85	−20.5 ± 8.65	36.3	+ 7.4
Inf. front. cortex	44.6 ± 2.35	−14.5 ± 3.97	39.4	+ 4.3
Sup. front. cortex	41.4 ± 1.99	−11.7 ± 6.92	30.0	+30.0
Premotor cortex	42.6 ± 1.83	−13.9 ± 4.81	38.4	+11.5
Sensorimot. cortex	44.2 ± 3.07	−14.2 ± 4.21	37.0	+16.2
Insular cortex	39.7 ± 1.39	−12.3 ± 3.36	30.8	+26.0
Inf. temp. cortex	35.4 ± 2.92	−10.6 ± 5.20	31.9	+11.6
Middle temp. cortex	38.8 ± 3.00	−11.8 ± 7.71	34.0	+ 8.8
Sup. temp. cortex	41.3 ± 2.11	−10.9 ± 6.97	36.8	+ 5.7
Inf. par. cortex	38.2 ± 2.68	−11.5 ± 5.40	31.0	+25.2
Sup. par. cortex	38.8 ± 1.75	−11.6 ± 4.57	33.7	+13.9
Vis. assoc. cortex	36.1 ± 3.07	−12.3 ± 4.98	29.1	+22.3
Prim. vis. cortex	40.1 ± 4.09	− 9.2 ± 9.84	35.5	+20.0
Inf. limbic struct.	32.3 ± 2.50	− 9.9 ± 4.76	27.9	+14.0
Hippocampus	38.3 ± 5.17	− 9.4 ± 9.33	31.6	+23.4
Cingulate gyrus	44.6 ± 2.86	−11.9 ± 7.62	41.0	+11.2
Front. bas. white matter	21.6 ± 0.83	−13.4 ± 5.14	18.6	+11.8
Occ. par. white matter	20.8 ± 1.72	−14.8 ± 1.88	17.0	+ 8.2
Caudate nucleus	40.5 ± 3.52	−11.6 ± 6.57	34.0	+ 7.9
Lentiform nucleus	41.6 ± 6.35	− 9.6 ± 8.85	38.3	+ 2.1
Thalamus	38.4 ± 3.21	−17.5 ± 7.22	31.0	+14.5
Brainstem	27.9 ± 4.85	− 6.8 ± 12.10	24.6	+17.1
Cerebellar gray matter	32.9 ± 5.96	− 9.5 ± 4.00	31.1	+ 9.3
Whole brain (all regions weighted according to size)	36.2 ± 2.38	−12.6 ± 4.73	30.3	+16.4

ten interactive program, which were then lumped into 23 topographically adjusted larger regions in each hemisphere according to Ward's method of cluster analysis, in order to minimize possible errors due to slightly different head positioning during repeat measurements.

Plasma glucose concentration (5.33 ± 0.538 μmol/liter, mean ± S.D.), pH (7.38 ± 0.023) and pCO_2 (42.0 ± 5.57 mm Hg) in arterialized venous blood were comparable in sleeping and waking subjects, while pO_2 decreased significantly ($P < 0.05$ by paired t-test) from 87.7 ± 7.27 mm Hg to 76.3 ± 3.30 mm Hg during sleep. In neither state a significant difference in regional cerebral metabolic rate for glucose (rCMRGl) could be detected between hemispheres and, therefore, data from corresponding regions were pooled. Although during wakefulness the dreaming volunteer's metabolic rates (weighted whole brain average = 30.3 μmol/100 g/min) differed significantly ($P < 0.01$ by analysis of variance a priori test) from the non-dreamers group (36.2 ± 2.38 μmol/100 g/min), all values still were well within the range found in other normal subjects of the same age group[3]. In every case the curve of dynamic FDG uptake by the brain recorded during the initial 30 min of study took a smooth course, thus indicating satisfactory metabolic steady state, i.e., despite several changes in sleep state any concomitant variation in glucose utilization did not exceed the natural scatter of radioactive decay. Both asleep and awake, in all subjects glucose metabolism differed significantly among regions ($P < 0.001$ by Greenhouse and Geisser's most conservative analysis of variance criterion for profile data[2]).

During sleep, as demonstrated in Table I and Fig. 1, the non-dreaming volunteers showed a significant ($P < 0.001$ by conservative two-way analysis of variance) generalized decrease in brain glucose utilization down to a weighted average of 31.6 ± 1.42 μmol/100 g/min for the whole brain. No region of a size large enough to be reproducibly detected by the PET system appeared activated, and differences in regional changes were not statistically significant (two-way ANOVA $P_{\text{interaction}} > 0.5$). The dreaming subject (Fig. 2), by contrast, not only exhibited considerably higher metabolic rates in all regions during his sleep study compared to wakefulness, but in some regions (e.g. superior frontal cortex, insula, inferior parietal cortex, hippocampus, and the entire visual cortex) increases were so far above the weighted whole brain average of +16.4% as to suggest major involvement of those structures in dream processing. (Because of the apparent interaction between brain regions and level of consciousness a Model I two-way analysis of variance without replication could not be properly applied.)

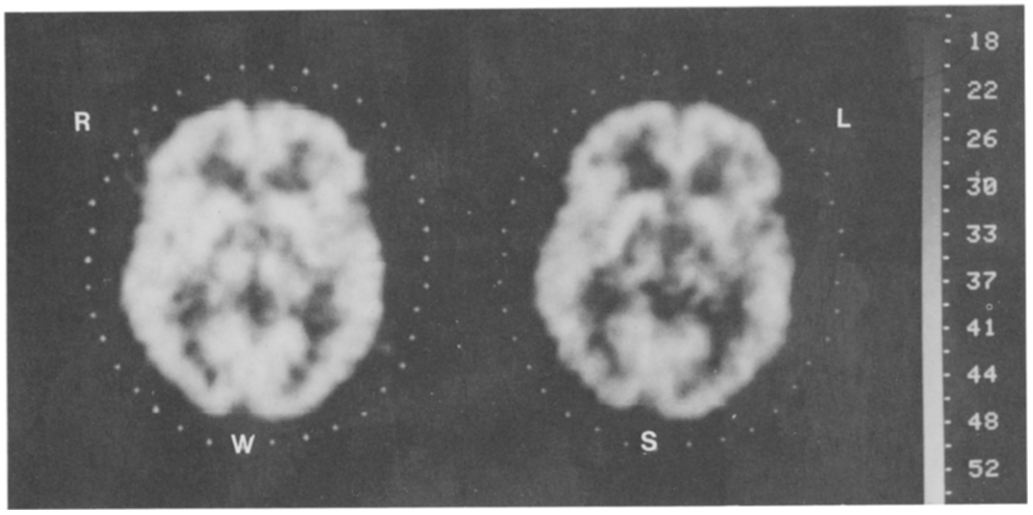


Fig. 1. Corresponding images of the local cerebral metabolic rate for glucose determined by FDG-PET in a characteristic horizontal brain slice across the basal ganglia of a healthy 37-year-old male subject representative of the non-dreamers group, showing non-selective decrease in glucose utilization from wakefulness (W) to sleep (S). Frontal poles are on top, occipital poles at bottom, sides as marked (R, L). Values on reference gray scale are in μmol/100 g/min.

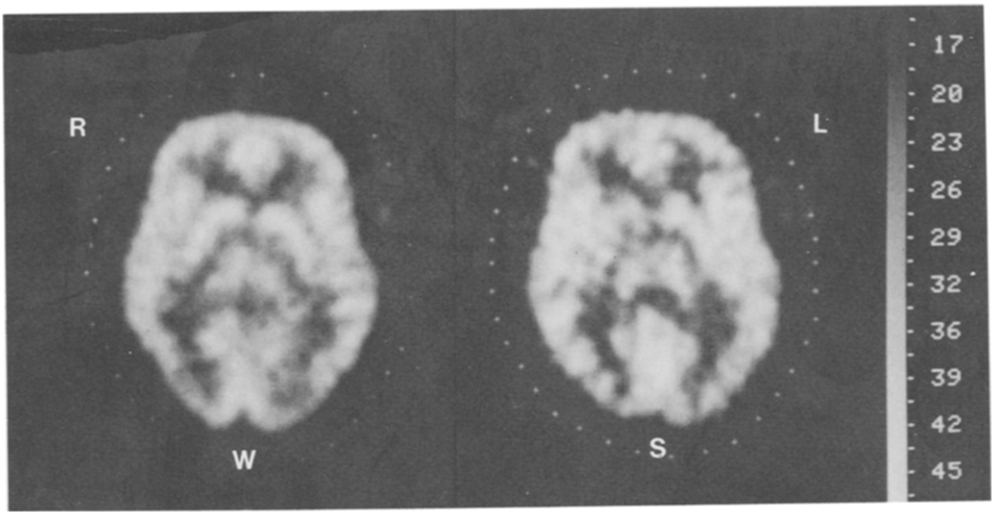


Fig. 2. Corresponding metabolic maps of a 28-year-old normal volunteer's brain slice recorded by FDG-PET while the subject was awake (W) and asleep dreaming (S). A generalized activation, most marked in the insular regions, visual cortex, and hippocampal formations, is clearly demonstrated during sleep with dreaming. Topography, orientation, and calibration are as described in the legend of Fig. 1.

The present results may shed some new light on an early study of Mangold et al.[6], who described an 11% decrease in global cerebral oxidative metabolism in 5 out of 6 normal young men during sleep, and an increase by 34% in the sixth volunteer, concluding that, overall, there was no significant change. In view of our observations those subjects may have represented two distinct sleeping states: non-dreaming and dreaming. The non-selective decrease in rCMRGl found in our group of non-dreaming volunteers also is in good agreement with results obtained by the 2-deoxy-D-[^{14}C]glucose autoradiographic method in rhesus monkeys during non-rapid eye movement (non-REM) sleep[5], while neither the differential effect of sleep on regional metabolic activity reported from semiquantitative autoradiographic experiments in rats[9], nor the predominant activation of cerebral and choroid plexus metabolism observed in cats during slow-wave sleep[7], are confirmed. Although in this pilot study eye movements were not specifically monitored, the metabolic changes found in dreaming and non-dreaming subjects are qualitatively consistent with the results of two-dimensional regional cerebral blood flow measurements in normal volunteers during REM and non-REM sleep[11,13], thereby re-emphasizing the coupling between cerebral metabolism and hemodynamics. It is hoped that these preliminary observations may spur further PET studies in humans, which eventually may provide deeper insight into the physiological enigma of sleep and dreaming.

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